

Jialong Wu

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EDUCATION

Ph.D. in Software Engineering

Aug 2022 – June 2027 (*expected*)

School of Software, Tsinghua University

Beijing, China

- Advisor: [Prof. Mingsheng Long](#)
- Research Interests: World Models, Reinforcement Learning

Bachelor in Software Engineering

Aug 2018 – June 2022

School of Software, Tsinghua University

Beijing, China

- GPA: 3.98/4.0, Rank: 1/89

Bachelor in Pure and Applied Mathematics (*Second Degree*)

Aug 2019 – June 2022

Department of Mathematical Sciences, Tsinghua University

Beijing, China

- GPA: 3.75/4.0

EXPERIENCE

Research Intern

March 2025 – *present*

ByteDance Seed

Beijing, China

PUBLICATIONS AND PREPRINTS

Visual Generation Unlocks Human-Like Reasoning through Multimodal World Models

Jialong Wu, Xiaoying Zhang[#], Hongyi Yuan, Xiangcheng Zhang, Tianhao Huang, Changjing He, Chaoyi Deng, Renrui Zhang, Youbin Wu, Mingsheng Long[#] *arXiv preprint*, 2026.

[[Project website](#)] [[PDF](#)] [[Code](#)] [[Datasets](#)]

Audio-Visual World Models: Towards Multisensory Imagination in Sight and Sound

Jiahua Wang^{*}, Shannan Yan^{*}, Leqi Zheng^{*}, Jialong Wu, Yaoxin Mao. *arXiv preprint*, 2025.

[[PDF](#)]

Vid2World: Crafting Video Diffusion Models to Interactive World Models

Siqiao Huang^{*}, Jialong Wu^{*}, Qixing Zhou, Shangchen Miao, Mingsheng Long[#]. *ICLR*, 2026.

[[Project website](#)] [[PDF](#)] [[Code](#)]

Foresight Diffusion: Improving Sampling Consistency in Predictive Diffusion Models

Yu Zhang^{*}, Xingzhuo Guo^{*}, Haoran Xu, Jialong Wu, Mingsheng Long[#]. *ICLR*, 2026.

[[PDF](#)]

RLVR-World: Training World Models with Reinforcement Learning

Jialong Wu, Shaofeng Yin, Ningya Feng, Mingsheng Long[#]. *NeurIPS*, 2025.

[[Project website](#)] [[PDF](#)] [[Code](#)] [[Models & Datasets](#)] [[Poster](#)]

Seed1.5-VL Technical Report

Dong Guo, Faming Wu, Feida Zhu, Fuxing Leng, Guang Shi, ... , Jialong Wu (*Contributor*), et al. *Technical report*, 2025.

[[Project website](#)] [[PDF](#)] [[Code](#)]

Trajectory World Models for Heterogeneous Environments

Shaofeng Yin^{*}, Jialong Wu^{*}, Siqiao Huang, Xingjian Su, Xu He, Jianye Hao, Mingsheng Long[#]. *ICML*, 2025.

[[PDF](#)] [[Code](#)]

CompilerDream: Learning a Compiler World Model for General Code Optimization

Chaoyi Deng^{*}, Jialong Wu^{*}, Ningya Feng, Jianmin Wang, Mingsheng Long[#]. *KDD*, 2025.

[[PDF](#)] [[Code](#)]

Long-Sequence Recommendation Models Need Decoupled Embeddings

Ningya Feng*, Junwei Pan*, **Jialong Wu***, Baixu Chen, Ximei Wang,
Qian Li, Xian Hu, Jie Jiang, Mingsheng Long#. *ICLR*, 2025.
[\[PDF\]](#) [\[Code\]](#)

iVideoGPT: Interactive VideoGPTs are Scalable World Models

Jialong Wu*, Shaofeng Yin*, Ningya Feng, Xu He, Dong Li, Jianye Hao, Mingsheng Long#. *NeurIPS*, 2024.
[\[Project Website\]](#) [\[PDF\]](#) [\[Code\]](#) [\[Models\]](#) [\[Poster\]](#) [\[Slides\]](#) [\[Blog\]](#)

HarmonyDream: Task Harmonization Inside World Models

Haoyu Ma*, **Jialong Wu***, Ningya Feng, Chenjun Xiao, Dong Li, Jianye Hao, Jianmin Wang, Mingsheng Long#. *ICML*, 2024.
[\[PDF\]](#) [\[Code\]](#) [\[Poster\]](#) [\[Slides\]](#) [\[Talk\]](#) [\[Blog\]](#)

Pre-training Contextualized World Models with In-the-wild Videos for Reinforcement Learning

Jialong Wu*, Haoyu Ma*, Chaoyi Deng, Mingsheng Long#. *NeurIPS*, 2023.
[\[PDF\]](#) [\[Code\]](#) [\[Poster\]](#) [\[Slides\]](#) [\[Blog\]](#)

CLIPood: Generalizing CLIP to Out-of-Distributions

Yang Shu*, Xingzhuo Guo*, **Jialong Wu**, Ximei Wang, Jianmin Wang, Mingsheng Long#. *ICML*, 2023.
[\[PDF\]](#) [\[Code\]](#)

Supported Policy Optimization for Offline Reinforcement Learning

Jialong Wu, Haixu Wu, Zihan Qiu, Jianmin Wang, Mingsheng Long#. *NeurIPS*, 2022.
[\[PDF\]](#) [\[Code\]](#) [\[Poster\]](#) [\[Slides\]](#)

Out-of-Dynamics Imitation Learning from Multimodal Demonstrations

Yiwen Qiu, **Jialong Wu**, Zhangjie Cao, Mingsheng Long#. *CoRL*, 2022.
[\[PDF\]](#) [\[Code\]](#)

Flowformer: Linearizing Transformers with Conservation Flows

Haixu Wu, **Jialong Wu**, Jiehui Xu, Jianmin Wang, Mingsheng Long#. *ICML*, 2022.
[\[PDF\]](#) [\[Code\]](#)

HONORS AND AWARDS

ByteDance Scholarship (awarded to 20 students in China and Singapore)	2025
ByteDance [Certificate]	
ICLR 2025 Notable Reviewer	2025
ICLR 2025 [Link]	
National Scholarship (Top 2%)	2020, 2024
Ministry of Education of the People's Republic of China [Certificate 2020] [Certificate 2024]	
Excellent Teaching Assistant Award	2024
Tsinghua University [Certificate]	
NeurIPS 2023 Top Reviewer	2023
NeurIPS 2023 [Link] [Certificate]	
Future Researcher Scholarship	2022
Tsinghua University [Certificate]	
Outstanding Graduates (Top 2%)	2022
School of Software [Certificate] & Tsinghua University [Certificate] & Beijing [Certificate]	
Excellent Graduation Thesis	2022
Tsinghua University [Certificate]	
Comprehensive Excellence Award (Top 10%)	2019, 2021, 2023, 2025
Tsinghua University [Certificate 2019] [Certificate 2021] [Certificate 2023]	
34th National Olympiad in Informatics (NOI 2017), Silver Medal (81st Place)	July 2017
China Computer Federation (CCF) [Certificate] [Link]	

INVITED TALKS AND ACTIVITIES

A Path Towards General World Models <u>BAAI Conference</u> , Beijing, China	June, 2025
Panelist <u>AIFC</u> Side Event of the <u>AI Action Summit 2025</u> , Paris, France	Feb, 2025
Towards General World Models: Pre-training, Multi-Modality, and Scalable Architecture ByteDance Research	June, 2024
Intrinsic Task Harmonization Inside World Models <u>RLChina</u>	Feb, 2024

PROFESSIONAL SERVICES

Reviewer | PC Member

- International Conference on Learning Representations (*ICLR*) 2024 – 2026
- Conference on Neural Information Processing Systems (*NeurIPS*) 2023 – 2025
- International Conference on Machine Learning (*ICML*) 2022 – 2026
- Conference on Computer Vision and Pattern Recognition (*CVPR*) 2026
- Transactions on Machine Learning Research (*TMLR*)

Teaching Assistant

- Deep Learning, Fall 2023, Tsinghua University. Instructor: Prof. Mingsheng Long.
- Introduction to Algorithms, Spring 2023, Tsinghua University. Instructor: Prof. Bin Wang.
- Machine Learning, Fall 2022, Tsinghua University. Instructor: Prof. Mingsheng Long.
- Introduction to Artificial Intelligence, Spring 2021, 2022, 2024, Tsinghua University. Instructor: Prof. Mingsheng Long.

Student Counselor

- Undergraduate Student Counselor, Dushi College, Tsinghua University, 2024–2027 (*expected*).

SELECTED PROJECTS

MiniVeo3-Reasoner: Thinking with Videos from Open-Source Priors 170+ Stars We reproduce chain-of-frames visual reasoning by fine-tuning open-source video models.	2025
Awesome Multi-task Learning 760+ Stars A curated list of datasets, codebases, and papers on Multi-Task Learning (MTL).	Up-to-date
ImageLabeler C++, Qt 120+ Stars, 40+ Forks Image annotation tool, supporting 2D/3D detection/segmentation annotation.	Summer 2019
Incremental Decision Tree Python (NumPy, Matplotlib, Pandas) Re-implementation of VFDT and EFDT, reproducing paper <i>Extremely Fast Decision Tree</i> .	Fall 2020

TECHNICAL SKILLS

Programming Language: Python, C/C++, MATLAB, Java, HTML/CSS, JavaScript
Professional Software: PyTorch, JAX, NumPy, Git, LaTeX, Vue
Language: China (native), English

Last updated: February 1, 2026